**Step 2:**

Also, more hours means less days. So, it is in inverse proportion again.

$\therefore$ The multiplying factor is $\frac{7}{8}$.

**Step 3:**

$\therefore x = 24 \times \frac{48}{28} \times \frac{7}{8} = 36$ days.

**Formula Method**

Here, $P_1 = 48$, $D_1 = 24$, $H_1 = 7$ and $W_1 = 1$ (Why?)

$P_2 = 28$, $D_2 = x$, $H_2 = 8$ and $W_2 = 1$ (Why?)

Using the formula, $\frac{P_1 \times D_1 \times H_1}{W_1} = \frac{P_2 \times D_2 \times H_2}{W_2}$

$$\Rightarrow \frac{48 \times 24 \times 7}{1} = \frac{28 \times x \times 8}{1}$$

$$\Rightarrow x = \frac{48 \times 24 \times 7}{28 \times 8} = 36 \text{ days.}$$



**Try these**

Identify the different variations present in the following questions:

1. 24 men can make 48 articles in 12 days. Then, 6 men can make _____ articles in 6 days.
2. 15 workers can lay a road of length 4 km in 4 hours. Then, _____ workers can lay a road of length 8 km in 8 hours.
3. 25 women working 12 hours a day can complete a work in 36 days. Then, 20 women must work _____ hours a day to complete the same work in 30 days.
4. In a camp there are 420 kg of rice sufficient for 98 persons for 45 days. The number of days that 60 kg of rice will last for 42 persons is _____.

## 4.6 Time and Work

How would you find the answer for the following question?

If Kani can finish a given work in 2 hours and Viji in 3 hours, then in what time can they finish it working together? The answer for this question will be known here in this section.

Work to be done is usually considered as one unit. Work can be in any form like building a wall, making a road, filling or emptying a tank or even eating a certain amount of food.

Time is measured in hours, days etc., Certain assumptions are made that the work so done is uniform and each person shares the same work time in case of group work in completing the work.



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